SMART INDIA HACKATHON 2026

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- **Problem Statement ID –** SIH26032
- **Problem Statement Title-** Farmers often face long waiting times, lack of information regarding procurement schedules, and uncertainty about procurement status.
- **Theme-** Ministry of Consumer Affairs, Food & Public Distribution
- **PS Category-** Software
- **Team ID-**
- **Team Name (Registered on portal) -** Synapse



IDEA TITLE

KisanSlot — Your Slot. Your Crop. Your Time.

❖ Proposed Solution — slot booking + live queue for MSP procurement centres

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HOW THE IDEA WORKS

- **Register once** — mobile OTP, land record, crop and expected quantity.
- **Book a slot** at the nearest centre — date, time band and gate, chosen from what is actually still open.
- **Get an SMS token** (e.g. KS-1042 · 11:00 AM · Gate 2) that works as the slot pass.
- **Live queue** — a public board and the farmer's phone both show "now serving" and his position.
- **One timeline per lot** — arrival, quality check, weighing, gate pass, MSP payment, each with an SMS.

HOW IT ADDRESSES THE PROBLEM

- **Waiting time:** arrivals spread across slots capped at the centre's real daily unloading capacity.
- **No schedule info:** centre-wise open dates, crops and remaining slots published before harvest.
- **Status uncertainty:** token-to-payment tracking replaces repeat visits and phone calls.

INNOVATION & UNIQUENESS

- **No smartphone needed** — SMS and IVR in the local language, plus CSC/VLE-assisted booking.
- **Capacity-aware slots** with a wait estimate from the queue's actual service rate.
- Adds the missing **live queue layer for wheat/paddy centres**.

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TECHNICAL APPROACH

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❖ Technologies to be used — a beginner-friendly, open-source stack

FRONTEND
HTML5, CSS3, JavaScript, Bootstrap 5 · responsive, works on a ₹5,000 phone browser

BACKEND
Python + Flask REST APIs · slot engine, token queue, role-based logins

DATABASE
SQLite for the demo, PostgreSQL for the pilot · farmers, centres, slots, lots, payments

COMMS & EXTRAS
MSG91 / Twilio SMS + IVR, Leaflet maps, Chart.js dashboards, QR gate pass

❖ Methodology and process for implementation — the farmer's journey, end to end

01 Register once — OTP, land record, crop → **02** Pick centre, date and time slot → **03** SMS token issued as the slot pass → **04** Arrive only at the booked time

05 Live queue — token calling and position → **06** Quality check and weighing entry → **07** e-Gate pass and digital receipt → **08** Payment status tracked till credit

6-member split: 2 frontend · 2 backend · 1 SMS/IVR + test data · 1 demo script, diagrams & docs

FEASIBILITY AND VIABILITY

❖ Analysis of the feasibility of the idea

- **Already proven by Government of India itself.** The Kapas Kisan App (Cotton Corporation of India) gives cotton farmers self-registration, slot booking, real-time updates on quality assessment and payment, and multilingual support — explicitly to cut paperwork, waiting time and distress sales. e-Samriddhi (NAFED) and e-Samyukti (NCCF) already issue **digitally scheduled slots** at a farmer's chosen purchase centre.
- **Digital rails and farmer readiness exist.** 1.80 crore farmers and 2.72 lakh traders are already registered on e-NAM across 1,656 mandis in 23 States and 4 UTs (as on 28 Feb 2026).
- **Buildable by a 6-member first-time team.** No AI, no hardware, no blockchain — a database, a booking rule, a queue counter and an SMS API. A demo for 2 centres and ~50 test farmers is realistic.

❖ Potential challenges and risks → ❖ Strategies for overcoming them

RISK / CHALLENGE	MITIGATION STRATEGY
Low smartphone / internet access	SMS + IVR in the local language as the primary channel; CSC / VLE and centre-desk assisted booking; one mobile number can book for a neighbour.
Fake or duplicate bookings blocking slots	OTP-verified mobile, one active booking per land record per crop, and auto-release of no-show slots back into the open pool.
Harvest-peak rush exceeds capacity	Slots capped at each centre's real daily unloading capacity, with a transparent waitlist and suggested alternate dates or nearby centres.
Staff adoption at the centre	A single-screen operator panel — call next token, mark weighed, mark paid; offline entry syncs when the network returns.

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IMPACT AND BENEFITS

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❖ Potential impact on the target audience

1.84 crore
farmers benefitted from MSP procurement in 2024-25, up from 1.63 crore in 2021-22

22 lakh+
wheat farmers served in RMS 2024-25, when FCI procured 266 LMT of wheat

₹3.33 lakh cr
MSP value disbursed to farmers in 2024-25 — each rupee tied to a visit and a payment cycle

297 LMT
procurement estimated for RMS 2026-27, worth about ₹84,263 crore at MSP

Source: PIB Backgrounder, Ministry of Consumer Affairs, Food & Public Distribution, 10 Oct 2025 — a queue-and-slot layer touches this footprint without changing MSP policy.

❖ Benefits of the solution (social, economic, environmental, etc.)

SOCIAL
No overnight waits at the gate with a loaded trolley. A booked time band lets a farmer plan his day, and elderly and women farmers are not forced into open-air queues.

ECONOMIC
Fewer wasted trips and less hired-vehicle idling. Visible payment status reduces pressure to sell below MSP to a private buyer just to get cash today.

ENVIRONMENTAL
Staggered arrivals cut tractor and truck idling and repeat trips, and grain spends less time lying in the open exposed to rain and spoilage.

ADMINISTRATIVE
Centre managers see expected arrivals a day ahead, so labour, weighbridges and gunny bags are planned instead of improvised — and every lot leaves an auditable trail.

RESEARCH AND REFERENCES

❖ Details / Links of the reference and research work

SOURCE	WHAT WE TOOK FROM IT	REFERENCE
PIB Backgrounder — "Stronger Procurement, Wider Farmer Coverage, Digital Reforms", 10 Oct 2025	Scale figures on slide 5 — 1.84 crore farmers benefitted, ₹3.33 lakh crore MSP value, 266 LMT wheat and 22 lakh farmers in RMS 2024-25, 297 LMT / ₹84,263 crore for RMS 2026-27. Also documents e-Samriddhi and e-Samyukti issuing *digitally scheduled slots*.	pib.gov.in — Release ID 2177219
Kapas Kisan App — Cotton Corporation of India, Ministry of Textiles	Closest existing precedent: self-registration, slot booking, real-time quality and payment updates, multilingual support — stated purpose is reducing paperwork, waiting time and distress sales. Confirms the approach is government-validated.	Documented in the same PIB Backgrounder, 10 Oct 2025
e-NAM status reply, Ministry of Agriculture & Farmers Welfare, Lok Sabha, 17 Mar 2026	Digital readiness evidence on slide 4 — 1,656 mandis across 23 States and 4 UTs, 1.80 crore farmers and 2.72 lakh traders registered as on 28 Feb 2026, 4,724 FPOs onboarded.	pib.gov.in — Release ID 2241414; enam.gov.in
NAFED e-Samridhi procurement portal	Reference flow for farmer registration with Aadhaar, land records and bank details, and for direct MSP payment — the data model our registration screen follows.	esamridhi.in
State procurement portals — Anaaj Kharid (Punjab), Meri Fasal Mera Byora (Haryana), HP APPP	Studied to locate the gap: these handle crop registration, land records, mandi schedules and payment tracking, but a farmer still has no per-token live queue position at the centre — the layer KisanSlot adds.	anaajkharid.in · fasal.haryana.gov.in · hpappp.hp.gov.in